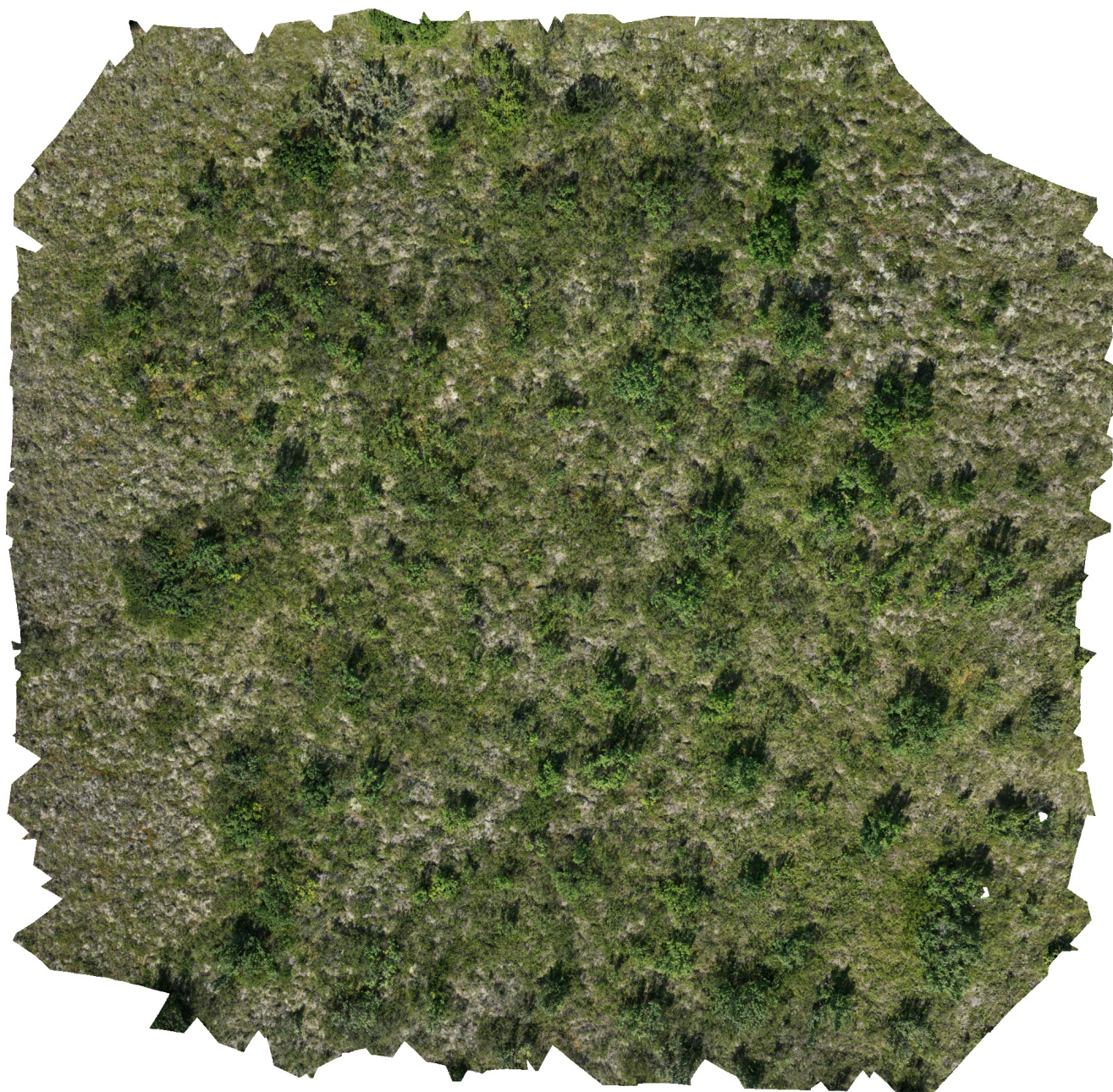


Agisoft Metashape

Processing Report

18 May 2024



Survey Data

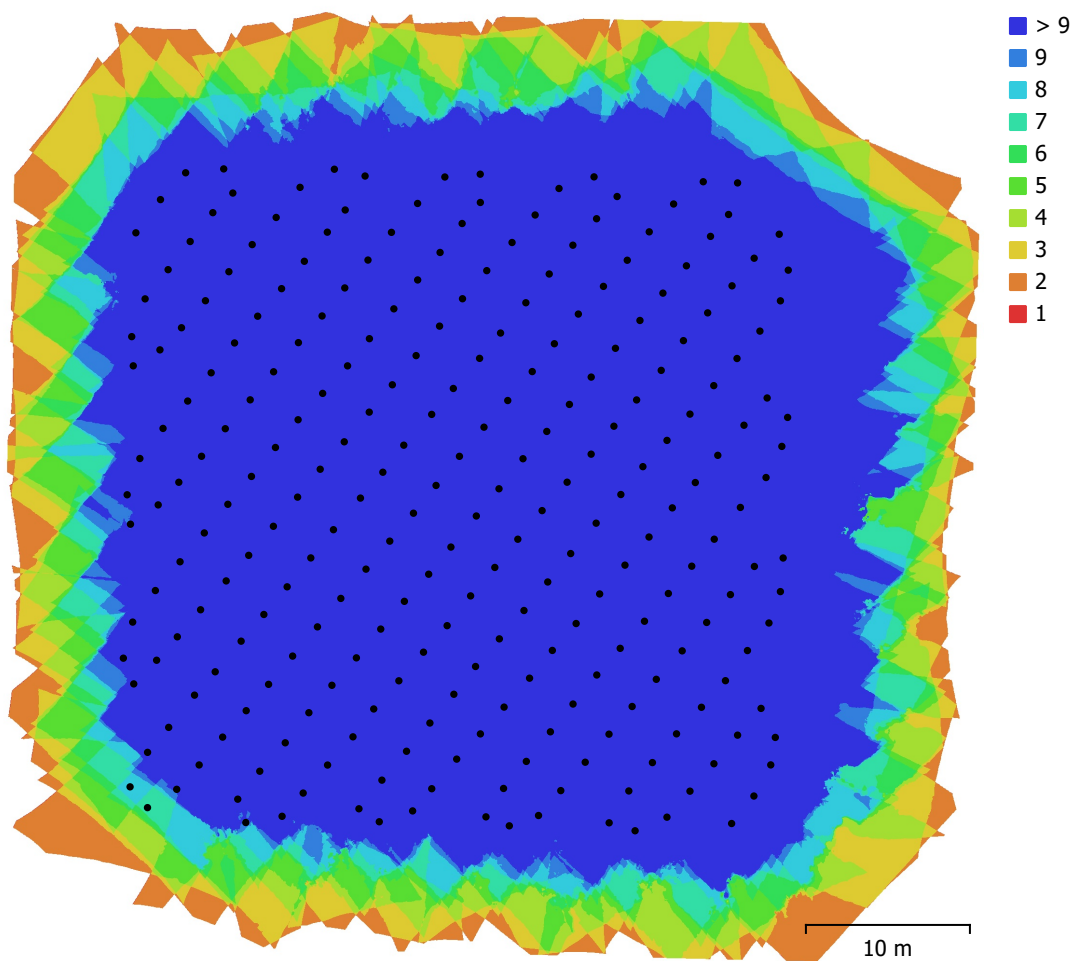


Fig. 1. Camera locations and image overlap.

Number of images:	258	Camera stations:	258
Flying altitude:	11 m	Tie points:	418,164
Ground resolution:	2.86 mm/pix	Projections:	1,099,472
Coverage area:	3.03e+03 m ²	Reprojection error:	1.01 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
M3M (12.29mm)	5280 x 3956	12.29 mm	3.36 x 3.36 μm	Yes

Table 1. Cameras.

Camera Calibration

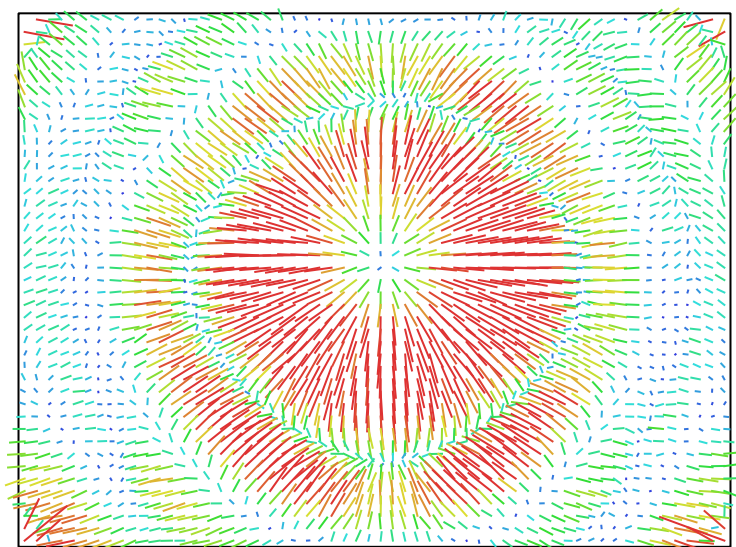


Fig. 2. Image residuals for M3M (12.29mm). 1 pix

M3M (12.29mm)

258 images, precalibrated

Type	Resolution	Focal Length	Pixel Size
Frame	5280 x 3956	12.29 mm	3.36 x 3.36 μm
F:	3727.67		
Cx:	20.7149	B1:	0
Cy:	0.241468	B2:	0
K1:	-0.12548	P1:	-6.53141e-05
K2:	0.0447886	P2:	9.38e-05
K3:	-0.0477832	P3:	0
K4:	0	P4:	0

Fixed parameters: f, cx, cy, k1, k2, k3, p1, p2

Camera Locations

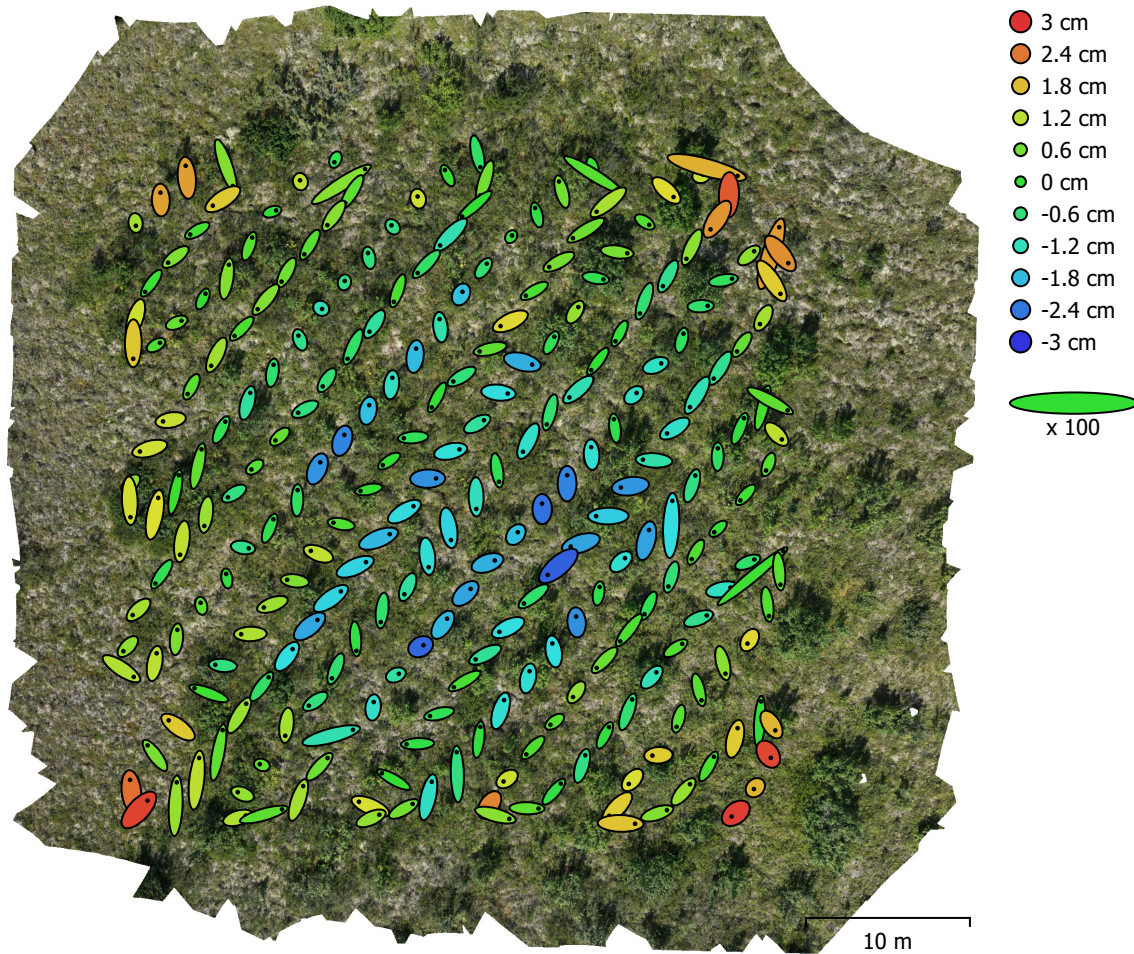


Fig. 3. Camera locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.

Estimated camera locations are marked with a black dot.

X error (cm)	Y error (cm)	Z error (cm)	XY error (cm)	Total error (cm)
0.883528	1.07183	1.13818	1.38905	1.7958

Table 2. Average camera location error.

X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

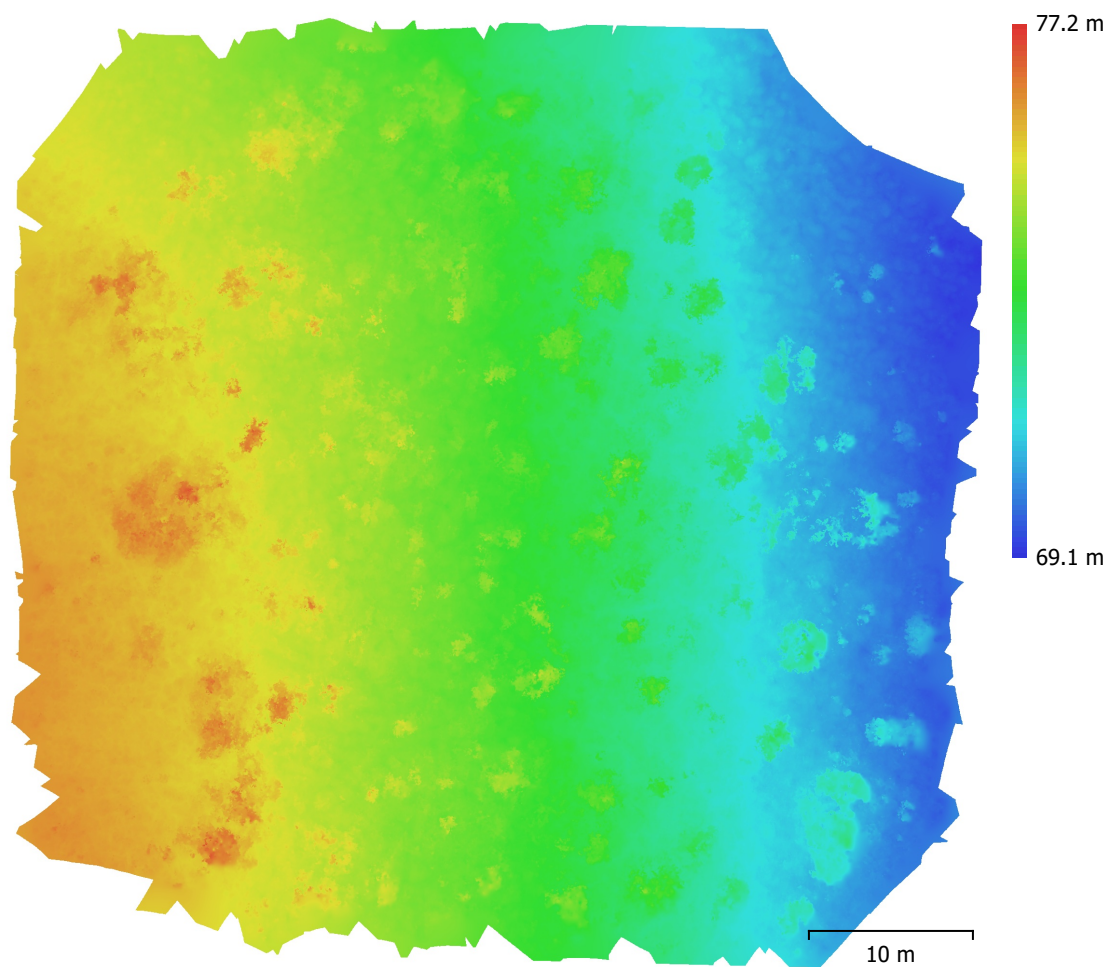


Fig. 4. Reconstructed digital elevation model.

Resolution: 5.73 mm/pix
Point density: 3.05 points/cm²

Processing Parameters

General

Cameras	258
Aligned cameras	258
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll

Tie Points

Points	418,164 of 433,785
RMS reprojection error	0.234104 (1.00593 pix)
Max reprojection error	0.705449 (78.3336 pix)
Mean key point size	3.64687 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	2.74776

Alignment parameters

Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	60,000
Key point limit per Mpx	1,000
Tie point limit	0
Exclude stationary tie points	No
Guided image matching	No
Adaptive camera model fitting	No
Matching time	7 minutes 7 seconds
Matching memory usage	659.74 MB
Alignment time	2 minutes 42 seconds
Alignment memory usage	158.11 MB
Date created	2024:05:10 19:49:44
Software version	2.1.0.17532
File size	29.03 MB

Depth Maps

Count	258
Depth maps generation parameters	
Quality	High
Filtering mode	Disabled
Max neighbors	16
Processing time	20 minutes 12 seconds
Memory usage	3.45 GB
Date created	2024:05:14 20:38:42
Software version	2.1.0.17532
File size	1.48 GB

Point Cloud

Points	123,388,764
--------	-------------

Point attributes

Color	3 bands, uint8
Normal	

Point classes

Created (never classified)	123,388,764
----------------------------	-------------

Depth maps generation parameters

Quality	High
---------	------

Filtering mode	Disabled
Max neighbors	16
Processing time	20 minutes 12 seconds
Memory usage	3.45 GB
Point cloud generation parameters	
Processing time	28 minutes 26 seconds
Memory usage	10.90 GB
Date created	2024:05:14 21:07:09
Software version	2.1.0.17532
File size	1.57 GB
DEM	
Size	10,274 x 9,966
Coordinate system	WGS 84 / UTM zone 8N (EPSG::32608)
Reconstruction parameters	
Source data	Point cloud
Interpolation	Enabled
Processing time	1 minutes 8 seconds
Memory usage	310.51 MB
Date created	2024:05:17 23:34:11
Software version	2.1.0.17532
File size	332.50 MB
Orthomosaic	
Size	20,523 x 19,891
Coordinate system	WGS 84 / UTM zone 8N (EPSG::32608)
Colors	3 bands, uint8
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	10 minutes 14 seconds
Memory usage	1.55 GB
Date created	2024:05:13 15:15:23
Software version	2.1.0.17532
File size	10.62 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.1.0 build 17532
OS	Windows 64 bit
RAM	127.68 GB
CPU	Intel(R) Core(TM) i9-9940X CPU @ 3.30GHz
GPU(s)	GeForce RTX 2080 Ti